Center Points

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Introduction

In this report we are going to apply center points to check if a second order model must be considered for the following database.

The Data

The data refer to two factors, we have a total of 8 responses in what appears to be an un-replicated design. We also include 5 center points

```
Y<-c(36,32,18,19,23,31,35,30) # the data
YC<-c(27,29,28,25,27) # the central points
```

We again create the design matrix as usual but this time we include the Y response values:

```
A   <- rep(c(-1,1), each = 4)
B   <- rep(rep(c(-1,1), each = 2), 2)
C   <- rep(rep(c(-1,1), each = 1), 4)
AB  <- A*B
AC  <- A*C
BC  <- C*B
ABC <- A*C*B
#We create the design matrix
Design <- data.frame(
  Intercept = 1,
  A   = A,
  B   = B,
  C   = C,
  AB  = AB,
  AC  = AC,
```

```r
  BC  = BC,
  ABC = ABC
)

Design_factorial <- data.frame(
  Y = Y,
  A = A, B = B, C = C,
  AB = AB, AC = AC, BC = BC, ABC = ABC
)
```

We also need to make the same matrix but with every A,B,C set to 0 for the center values:

```r
Design_center <- data.frame(
  Y = YC,
  A = 0, B = 0, C = 0,
  AB = 0, AC = 0, BC = 0, ABC = 0
)
```

We now combine them to create a common design matrix while also labeling which should be considered part of the regular ANOVA process(factorial) and which are the center ones:

```r
Design_all <- rbind(Design_factorial, Design_center)
Design_all$type <- ifelse(Design_all$A == 0 & Design_all$B == 0 & Design_all$C == 0, "Center"

Design_all
```

```
    Y  A  B  C AB AC BC ABC       type
1  36 -1 -1 -1  1  1  1  -1 Factorial
2  32 -1 -1  1  1 -1 -1   1 Factorial
3  18 -1  1 -1 -1  1 -1   1 Factorial
4  19 -1  1  1 -1 -1  1  -1 Factorial
5  23  1 -1 -1 -1 -1  1   1 Factorial
6  31  1 -1  1 -1  1 -1  -1 Factorial
7  35  1  1 -1  1 -1 -1  -1 Factorial
8  30  1  1  1  1  1  1   1 Factorial
9  27  0  0  0  0  0  0   0    Center
10 29  0  0  0  0  0  0   0    Center
11 28  0  0  0  0  0  0   0    Center
12 25  0  0  0  0  0  0   0    Center
13 27  0  0  0  0  0  0   0    Center
```

Now we will attempt to do the Three-Way ANOVA as usual:

```r
model_fact <- lm(Y ~ A * B * C, data = subset(Design_all, type == "Factorial"))
summary(model_fact)
```

```
Call:
lm(formula = Y ~ A * B * C, data = subset(Design_all, type ==
    "Factorial"))

Residuals:
ALL 8 residuals are 0: no residual degrees of freedom!

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.800e+01        NaN     NaN      NaN
A            1.750e+00        NaN     NaN      NaN
B           -2.500e+00        NaN     NaN      NaN
C            3.592e-15        NaN     NaN      NaN
A:B          5.250e+00        NaN     NaN      NaN
A:C          7.500e-01        NaN     NaN      NaN
B:C         -1.000e+00        NaN     NaN      NaN
A:B:C       -2.250e+00        NaN     NaN      NaN

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared:      1, Adjusted R-squared:    NaN
F-statistic:   NaN on 7 and 0 DF,  p-value: NA
```

We will now include the curvature in the ANOVA, this is done by adding a dummy variable to the design matrix, which equals to one for center values and 0 for the factorial ones:

```r
Design_all$Curv <- ifelse(Design_all$type == "Center", 1, 0)

model_curv <- lm(Y ~ A*B*C + Curv, data = Design_all)
summary(model_curv)
```

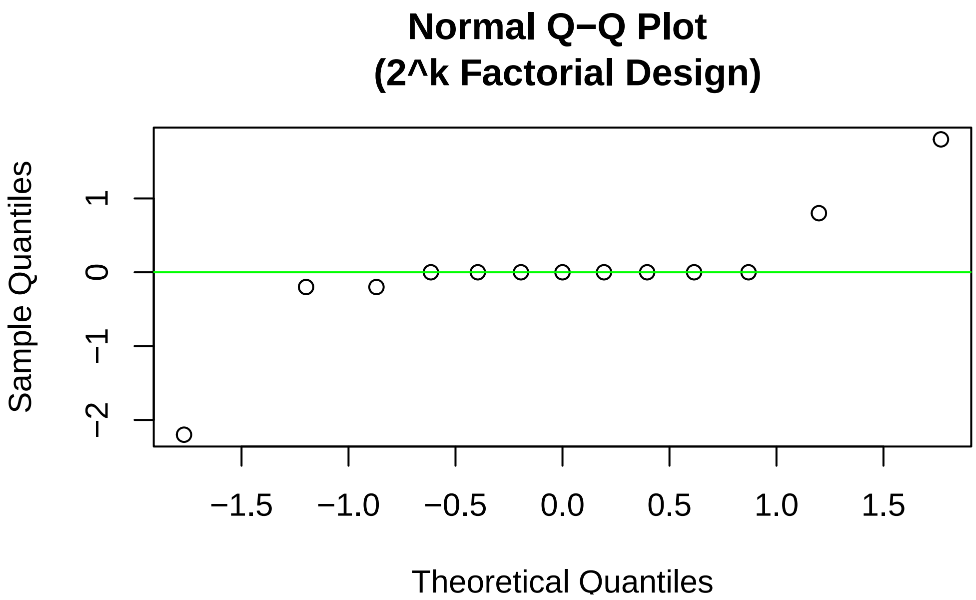
```
Call:
lm(formula = Y ~ A * B * C + Curv, data = Design_all)

Residuals:
```

```
         1           2           3           4           5           6           7 
-6.939e-18  1.931e-17 -2.822e-17  3.784e-17  6.194e-18  2.365e-17 -4.036e-18 
         8           9          10          11          12          13 
-1.016e-17 -2.000e-01  1.800e+00  8.000e-01 -2.200e+00 -2.000e-01 

Coefficients:
              Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2.800e+01  5.244e-01  53.394 7.36e-07 ***
A            1.750e+00  5.244e-01   3.337  0.02891 *  
B           -2.500e+00  5.244e-01  -4.767  0.00886 ** 
C           -2.854e-15  5.244e-01   0.000  1.00000    
Curv        -8.000e-01  8.456e-01  -0.946  0.39767    
A:B          5.250e+00  5.244e-01  10.011  0.00056 ***
A:C          7.500e-01  5.244e-01   1.430  0.22589    
B:C         -1.000e+00  5.244e-01  -1.907  0.12920    
A:B:C       -2.250e+00  5.244e-01  -4.291  0.01274 *  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.483 on 4 degrees of freedom
Multiple R-squared:  0.9755,	Adjusted R-squared:  0.9264 
F-statistic: 19.88 on 8 and 4 DF,  p-value: 0.005727
```

It is now clear that the curvature is not significant, thus the linear model is sufficient for our analysis.

Let's also do the Normal Probability Plot:

```r
qqnorm(residuals(model_curv),main = "Normal Q-Q Plot \n (2^k Factorial Design)")
qqline(residuals(model_curv),col="green")
```



Well the errors are the same everywhere with slight changes in curvature etc. These are all created from the replication of the center values.